Paola Jayme M. Romaguera

Spatial Analyst | GIS and Remote Sensing Specialist | Climate Risk Consultant

PROFESSIONAL SUMMARY

Climate risk and geospatial analytics specialist with 5+ years of experience leading end-to-end climate risk assessments and multidisciplinary hazard modeling initiatives. Demonstrated leadership in mentoring junior analysts, developing and standardizing internal modeling methodologies, and optimizing geospatial workflows for scalable asset-level risk analysis under multiple climate scenarios (TCFD-aligned). Combines advanced spatial analytics expertise with strong stakeholder engagement and business development experience.

WORK EXPERIENCE

SyCip Gorres Velayo & Co. (SGV, a member firm of Ernst and Young)

Senior Associate – Climate Change and Sustainability Services (January 2024 – Present)

Climate Risk Modeling & Assessment

- Led TCFD-aligned climate risk assessments for 500+ assets, quantifying exposure to floods, landslides, and extreme rainfall under multiple RCP scenarios.
- Designed hazard models (hydrodynamic flood simulations; slope-stability analyses) using Python (NumPy, xarray, rasterio) and QGIS to generate forward-looking risk maps.
- Automated Google Earth Engine workflows to assess turbidity and land-water change using remote-sensing indices, supporting climate-risk interpretation and exposure mapping.

Business Development & Client Engagement

- Drafted and co-authored technical proposals that secured three new climate risk assessment engagements within one year, including a major conglomerate and an electronics manufacturing company.
- Facilitated vulnerability & exposure workshops with asset owners, aligning risk metrics to business impacts and guiding adaptation strategies.

Targeted Collaborations

- Conducted climate-risk assessment for an agricultural company with operations in the Philippines and Indonesia, integrating crop-sensitivity models with temperature and precipitation anomaly analyses to guide agriculturists and other experts in their studies.
- Performed hydrometric and river-height analyses for a government agency's feasibility study, informing infrastructure siting and design.

Reporting & Communication

- Produced executive-level risk dashboards, technical reports, and slide decks in Word and PowerPoint; translated complex model outputs into clear recommendations for senior management.

Philippine Institute of Volcanology and Seismology (PHIVOLCS)

Science Research Specialist (March 2023 – December 2023)

- Developed integrated geospatial workflows to assess seismic and secondary-hazard risk, cutting data-processing time by 50%.
- Built Python scripts for spatial hazard-exposure extraction, automating multi-layer overlay analyses in ArcGIS/QGIS.
- Synthesized field and remote-sensing data into risk-assessment deliverables for national hazard mapping.

University of the Philippines – Resilience Institute (UPRI)

Geographic Information System (GIS) Researcher and Operator (August 2020 – February 2023)

- Supported scenario-based flood and volcanic hazard modeling, processing large raster datasets for baseline and future-climate runs.
- Automated Ash3d volcanic dispersion simulations via Bash and Python for rapid multi-scenario outputs.
- Authored scientific briefs in LaTeX to communicate hazard-risk results to multidisciplinary teams.

EDUCATION

University of the Philippines

Bachelor of Science in Geology

SKILLS

- **Climate Risk Modeling:** Hydrodynamic flood simulation (Flo-2d), slope-stability (TRIGRS), multi-scenario exposure analysis
- **Programming & Automation:** Python (NumPy, Pandas, xarray, rasterio, geopandas), Bash scripting, Google Earth Engine
- **GIS & Remote Sensing:** ArcGIS, QGIS, satellite-derived indices, spatial analysis
- **Frameworks & Reporting:** TCFD alignment, risk-mapping, stakeholder workshops
- **Tools & Platforms:** Microsoft Office (Word, Excel, PowerPoint), LaTeX, Git/GitHub, Streamlit, Tableau

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